

October 31, 2025 11:43 PM GMT

## Global Macro Commentary | Global

## October 31: Fedspeak Friday

Fed's Logan and Hammack lean hawkish; US President Trump denies potential military action within Venezuela; upside surprise in Japan inflation; continued budget challenges in France; Colombia BanRep on hold, as expected; DXY at 99.804 (+0.3%); US 10y at 4.077 (-2.0bp).

**Global rates see relatively muted moves despite a host of statements from Fed policymakers and noise on the geopolitical front.**

*Developed Markets*

- **US rates see mixed moves as the day's Fedspeak leans hawkish, but rally into the close with potential impacts from month-end index extensions.** In the New York morning, risk sentiment weakens on the back of speculative reports that the US is considering military action on Venezuelan facilities. President Trump later [refutes](#) the reports, but geopolitical concerns keep Treasury yields fairly low even as regional Fed presidents Logan and Hammack — both of whom will be voting FOMC members in 2026 — push back against the recent October rate cut, saying they would have preferred that the Fed remain on hold.
- **The front end outperforms the long end in the initial risk-off move amid rising inflation expectations.** 10y breakevens widen 2bp and remain wide even as oil prices retrace lower after President Trump's comments. Front-end Treasuries rally into the close as Fed Governor Waller calls for a rate cut in December and states that he would accept the role of Fed Chair if asked. As a result, the curve ultimately bull-steepens (2y: -4bp; 30y: -0bp).
- **In her speech, Logan also calls for the Fed to begin asset purchases if repo rates remain elevated.** Afterwards, swap spreads widen, led by the front end, which closes ~2bp wider. In light of Logan's comments, our rates strategists anticipate that primary dealers will be more willing use the standing repo facility to obtain funding when it is economical to do so.
- **The dollar continues to strengthen against G10 currencies (DXY: +0.3%).** CAD stores fewer losses against USD (USD/CAD: +0.2%) than the rest of the bloc, despite President Trump [refusing](#) to restart trade talks with Canada following last week's disagreement. Meanwhile, a US court ruling directs the administration to provide funding for federal food assistance, which could reduce some of the [consumer-driven economic risks](#) associated with the government shutdown.
- **Japanese rates twist-steepen, with the front end outperforming after a strong auction while the long end continues to sell off alongside the**

MORGAN STANLEY &amp; CO. LLC

**Bradley Tian**

Strategist

Bradley.Tian@morganstanley.com

+1 212 761-0826

**Molly Nickolin**

Strategist

Molly.Nickolin@morganstanley.com

+1 212 761-3592

**Lingdi Xu**

Economist

Lingdi.Xu@morganstanley.com

+1 212 761-2957

**Sofia Palacios**

Strategist

Sofia.Palacios@morganstanley.com

+1 212 761-0428

MORGAN STANLEY ASIA LIMITED+

**Gek Teng Khoo**

Strategist

Gek.Teng.Khoo@morganstanley.com

+852 3963-0303

MORGAN STANLEY MUFG SECURITIES CO., LTD. +

**Hiromu Uezato**

Strategist

Hiromu.Uezato@morganstanleymufg.com

+81 3 6836-8431

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

**Nimish M Prabhune**

Strategist

Nimish.Prabhune@morganstanley.com

+91 22 6996-1862

MORGAN STANLEY &amp; CO. INTERNATIONAL PLC+

**Maria Chiara Russo**

Strategist

Maria.Chiara.Russo@morganstanley.com

+44 20 7677-3499

**David Cueva**

Strategist

David.Cueva@morganstanley.com

+44 20 7425-2143

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

**For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.**

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

**overnight move in Treasuries.** The 2y JGB auction comes in slightly strong with a higher-than-expected cut-off price of 100.13 (C: 100.125), driving a rally in the Tokyo afternoon (2y: -1bp). JPY initially strengthens on an upside surprise in Tokyo CPI (2.8%; C: 2.6%) and a verbal intervention by Finance Minister Katayama, but ultimately retraces to end the New York session little changed (USD/JPY: -0.1%).

- **Euro area rates twist-steepen modestly (2y: -2bp; 30y: +1bp), with political developments in France driving weakness in the long end.** The OAT/Bund spread widens as negotiations around the budget continue, with the parties ultimately [failing to reach a compromise](#) on the proposed wealth tax. The 30y BTP/OAT spread touches 0bp intraday, a level not seen since 1998. Regional October inflation data comes in roughly in line with expectations at both the country and aggregate level, and the EUR swaps market also sees steepening pressure.

#### Emerging Markets

- **After Colombia's BanRep holds rates at 9.25%, rates sell off across the IBR curve and COP strengthens 0.2%, suggesting market participants view the messaging as hawkish.** Our economists [believe](#) the persistence of inflation and fiscal uncertainty have raised the bar for renewed easing. They look towards the BanRep minutes to be released Monday, as well as any potential upward revision in BanRep's inflation forecast in the Q3 inflation report. Elsewhere in LatAm, Mexico's TIE curve trades flat, mirroring the move in US rates.
- **AxJ currencies and rates also see mixed but modest moves, with Korea outperforming.** After September industrial production comes in higher than expected at 11.6% y/y (C: 5.9%), KRW climbs 0.2% against USD alongside a rally in local equities (KOSPI: +0.5%). Local rates also outperform despite the strong print — the curve bull-steepens (1y1y: -7bp) in a reversal of Thursday's sharp sell-off.
- **In China, rates rally 1-2bp in a risk-off move following a downside surprise in October manufacturing PMI (49.0; C: 49.6), with local equities selling off (CSI 300: -1.5%).** Meanwhile, Indian rates rally (10y IGB: -4bp) after the RBI sells fewer bonds than planned (INR210bn vs. INR320bn expected). INR weakens slightly against USD amid unconfirmed [reports](#) of RBI intervention in the FX market.
- **CEEMEA rates see mixed performance, with Poland outperforming on the back of a downside surprise in CPI at 2.8% (C: 3.0%).** Rates rally across the curve, with 2y yields falling 4bp. PLN closes 0.2% weaker against EUR and 0.5% weaker against USD. The rest of the region sees a steepening pressure in line with core rates, with twist-steepening in South Africa (2y: -1bp; 10y: +2bp) and Turkey (OIS 2y: -4bp; 10y: +5bp) along with bear-steepening in the Czech Republic (2y: +2bp; 10y: +4bp).

## Central Bank Monitor

### Developed Markets

- **Dallas Fed President Logan** emphasized a cautious stance on monetary policy, saying she would have preferred to hold rates steady this week given still-high inflation and uncertainty. She noted that inflation is “not convincingly headed” back to the 2% target, reiterated the need to modernize the target rate framework, and highlighted that while balance sheet runoff has improved efficiency, liquidity risks remain.
- **Cleveland Fed President Hammack** pushed back against the October rate cut, saying she would have preferred to hold rates steady. She stressed that policy is “barely restrictive, if at all,” and argued for staying restrictive to bring inflation down, noting inflation is broader than tariffs and core services remain strong. While acknowledging some signs of labor softness, she emphasized the need to maintain a robust picture of the economy and remain open to future adjustments.
- **Fed Governor Waller** [reiterated](#) a dovish view, suggesting the Fed should continue cutting rates in December, citing risks of labor market weakness.

### Emerging Markets

- In Colombia, **BanRep maintained its policy** rate at 9.25% with a vote split of 4 in favor of no change, 1 in favor of a 25bp cut, and 2 in favor of a 50 bp cut. The statement flagged robust domestic demand, widening trade deficit, and easier external conditions, while warning of upside risks from fiscal pressures and the minimum wage hike. In the press conference, Governor Villar mentioned that Colombia’s monetary policy stance remains contractionary, with the baseline scenario still pointing to eventual easing rather than hikes, though less likely scenarios could include rate increases if inflation risks materialize. Inflation in 2026 is expected to reach 4.25%, with cuts expected to resume early 2026, ending the year at 8.25%. [Read more](#)
- In China, **PBoC Governor Pan** [said](#) the central bank will enhance the base money supply mechanism, maintain reasonable growth in aggregate finance, and improve the formation, regulation, and transmission of market-based interest rates. Regarding financial stability, he noted the need to correct herd effect in the markets in a timely manner and analyze the toolkit for managing macroprudential and financial market volatility.

- In Indonesia, **BI Governor Warjiyo** affirmed the central bank's plan to develop the official national version of the stablecoin, which is "a digital version of the BI IDR security, a digital IDR with government bonds as underlying."

**Exhibit 1:** G4 Rates Closes (5pm NY)

| Tenor | US     |           |     | Germany |           |     | UK     |           |     | Japan  |           |     |
|-------|--------|-----------|-----|---------|-----------|-----|--------|-----------|-----|--------|-----------|-----|
|       | 31-Oct | 1d Δ (bp) | 2wk | 31-Oct  | 1d Δ (bp) | 2wk | 31-Oct | 1d Δ (bp) | 2wk | 31-Oct | 1d Δ (bp) | 2wk |
| 2y    | 3.574  | (3.5)     |     | 1.965   | (1.9)     |     | 3.765  | (2.5)     |     | 0.919  | (0.8)     |     |
| 3y    | 3.578  | (3.1)     |     | 2.023   | (2.1)     |     | 3.763  | (2.2)     |     | 0.996  | (0.1)     |     |
| 5y    | 3.687  | (3.1)     |     | 2.231   | (2.0)     |     | 3.881  | (2.6)     |     | 1.225  | 1.4       |     |
| 7y    | 3.871  | (2.8)     |     | 2.370   | (1.6)     |     | 4.022  | (1.7)     |     | 1.453  | 1.8       |     |
| 10y   | 4.078  | (2.0)     |     | 2.632   | (1.0)     |     | 4.408  | (1.4)     |     | 1.663  | 1.6       |     |
| 20y   | 4.627  | (0.6)     |     | 3.107   | 0.4       |     | 5.063  | (0.7)     |     | 2.592  | 1.7       |     |
| 30y   | 4.651  | (0.3)     |     | 3.211   | 1.3       |     | 5.177  | (0.3)     |     | 3.051  | 1.5       |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 2:** Inflation Closes (5pm NY)

| US TIPS |        |           |     | 10y Real Yields, BEI |        |           |     |      |        |           |     |
|---------|--------|-----------|-----|----------------------|--------|-----------|-----|------|--------|-----------|-----|
| Tenor   | 31-Oct | 1d Δ (bp) | 2wk | 10y                  | 31-Oct | 1d Δ (bp) | 2wk | 10y  | 31-Oct | 1d Δ (bp) | 2wk |
| 5y      | 1.270  | (6.1)     |     | DBRi                 | 0.665  | (1.8)     |     | BTPi | 1.678  | (1.7)     |     |
| BEI     | 2.411  | 2.8       |     | BEI                  | 1.758  | 0.2       |     | BEI  | 1.730  | 0.4       |     |
| 10y     | 1.759  | (4.0)     |     | UKTi                 | 1.548  | (2.2)     |     |      |        |           |     |
| BEI     | 2.316  | 1.9       |     | BEI                  | 2.927  | 0.7       |     |      |        |           |     |
| 30y     | 2.414  | (1.8)     |     | JGBi                 | 0.042  | 0.5       |     |      |        |           |     |
| RFI     | 2.241  | 1.4       |     | RFI                  | 1.590  | 1.0       |     |      |        |           |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 3:** Macro Closes (5pm NY)

| Spot | Majors |          |     | Spot | EM      |          |     | Index     | Equities / Commodities |          |     |
|------|--------|----------|-----|------|---------|----------|-----|-----------|------------------------|----------|-----|
|      | 31-Oct | 1d Δ (%) | 2wk |      | 31-Oct  | 1d Δ (%) | 2wk |           | 31-Oct                 | 1d Δ (%) | 2wk |
| DX   | 99.80  | 0.3      |     | CNH  | 7.12    | 0.2      |     | S&P       | 6840.20                | 0.3      |     |
| EUR  | 1.154  | (0.2)    |     | INR  | 88.77   | 0.1      |     | Stoxx     | 5662.04                | (0.7)    |     |
| GBP  | 1.315  | 0.0      |     | THB  | 32.34   | (0.1)    |     | FTSE      | 9717.25                | (0.4)    |     |
| JPY  | 153.99 | (0.1)    |     | KRW  | 1430.00 | (0.0)    |     | Nikkei    | 52411.34               | 2.1      |     |
| CHF  | 0.805  | 0.3      |     | ZAR  | 17.33   | 0.3      |     | VIX       | 17.44                  | 3.1      |     |
| AUD  | 0.655  | (0.2)    |     | MXN  | 18.55   | 0.1      |     | Gold (S)  | 4002.92                | (0.5)    |     |
| CAD  | 1.401  | 0.2      |     | BRL  | 5.38    | (0.1)    |     | WTI (Fut) | 60.98                  | 0.7      |     |

Source: Morgan Stanley Research, Bloomberg

**Exhibit 4:** G4 Central Bank Implied Pricing

| Fed          | Central Bank OIS |           |         |  | ECB           | Central Bank OIS |           |         |  | BOE           | Central Bank OIS |           |         |  | BOJ           | Central Bank OIS |           |         |  |
|--------------|------------------|-----------|---------|--|---------------|------------------|-----------|---------|--|---------------|------------------|-----------|---------|--|---------------|------------------|-----------|---------|--|
|              | Implied          | 1d Δ (bp) | Total Δ |  |               | Implied          | 1d Δ (bp) | Total Δ |  |               | Implied          | 1d Δ (bp) | Total Δ |  |               | Implied          | 1d Δ (bp) | Total Δ |  |
| EFFR (10/30) | 3.870            | 0.0       | —       |  | EOBIA (10/30) | 1.931            | 0.0       | —       |  | SONIA (10/30) | 3.969            | 0.0       | —       |  | TONAR (10/30) | 0.476            | 0.0       | —       |  |
| 12/10/25     | 3.712            | 1.1       | (15.8)  |  | 12/18/25      | 1.917            | 0.0       | (1.4)   |  | 11/06/25      | 3.896            | (0.9)     | (7.3)   |  | 12/19/25      | 0.596            | 0.5       | 12.0    |  |
| 01/28/26     | 3.628            | 1.0       | (24.2)  |  | 02/05/26      | 1.900            | 0.1       | (3.1)   |  | 12/18/25      | 3.799            | (1.3)     | (17.0)  |  | 01/23/26      | 0.690            | 0.5       | 21.4    |  |
| 03/18/26     | 3.533            | 0.8       | (33.7)  |  | 03/19/26      | 1.854            | (0.3)     | (7.7)   |  | 02/05/26      | 3.649            | (0.9)     | (32.0)  |  |               |                  |           |         |  |
| 04/29/26     | 3.473            | 0.5       | (39.7)  |  | 04/30/26      | 1.847            | (0.6)     | (8.4)   |  | 03/19/26      | 3.591            | (0.9)     | (37.9)  |  |               |                  |           |         |  |
| 06/17/26     | 3.320            | (0.7)     | (55.0)  |  | 06/11/26      | 1.813            | (1.0)     | (11.8)  |  | 04/30/26      | 3.480            | (0.8)     | (49.0)  |  |               |                  |           |         |  |
| 07/29/26     | 3.233            | (1.0)     | (63.7)  |  | 07/22/26      | 1.809            | (1.0)     | (12.2)  |  | 06/18/26      | 3.442            | (1.5)     | (52.7)  |  |               |                  |           |         |  |

Source: Morgan Stanley Research, Bloomberg

**Economic Releases***Developed Markets*

- Japan October Tokyo Core CPI:** 2.8% y/y (C: 2.6%, P: 2.5%) & **Core-of-core CPI (excluding fresh food and energy):** 2.8% y/y (C: 2.6%, P: 2.5%). The upside surprise came from fading out of Tokyo government's free basic water charges program, which pushed up the y/y CPI growth by 25bp. While our economists see the trend in underlying inflation as little changed, they

maintain their view on December hike. [Read more](#)

- **Netherlands October Flash CPI:** 3.1% (MSe: 3.2%; C: 3.1%; P: 3.3%)
- **France October Flash CPI:** 1.0% (MSe: 1.1%; C: 1.0%; P: 1.2%)
- **Italy October Flash CPI:** 1.2% (MSe: 1.4%; C: 1.5%; P: 1.6%)
- **Euro Area October Flash HICP:** 2.1% (MSe: 2.2%; C: 2.1%; P: 2.2%); **Core HICP:** 2.4% (MSe: 2.4%; C: 2.3%; P: 2.4%)
- **Portugal October Flash CPI:** 2.3% (MSe: 2.3%; P: 2.4%)
- **Canada August Preliminary GDP:** -0.3% m/m (C: 0.0%; P: 0.9%). Monthly industry-level GDP data showed the Canadian economy contracted by nearly 0.3% in August. The decline in economic activity was due in large part to a contraction in trade-dependent industries.

### Emerging Markets

- **Korea September Industrial Production:** 11.6% y/y (C: 5.9%; P: 0.7%)
- **China October NBS Manufacturing PMI:** 49.0 (MSe: 49.6; C: 49.6; P: 49.8); **NBS Non-Manufacturing PMI:** 50.1 (C: 50.1; P: 50.0); **NBS Composite PMI:** 50.0 (P: 50.6)
- **Singapore September M2 Money Supply:** 8.5% y/y (P: 9.5%)
- **Thailand September Current Account Balance:** \$1.9bn (C: \$500mn; P: -\$1.5bn); **September Trade Balance:** \$3.6bn (P: \$810mn); **September Exports:** 19.2% y/y (P: 5.5%) and **Imports:** 18.0% y/y (P: 14.7%)
- **Taiwan 3Q25 Advance GDP:** 7.6% y/y (MSe: 5.5%; C: 6.3%; P: 8.0%)
- **Hong Kong 3Q25 Advance GDP:** 3.8% y/y (C: 3.0%; P: 3.1%); **September Retail Sales Value:** 5.9% y/y (C: 2.6%; P: 3.9%); **Retail Sales Volume:** 4.8% y/y (C: 1.9%; P: 3.4%)
- **India September Fiscal Deficit YTD:** INR5731bn (P: INR5982bn)
- **Turkey September Trade Balance:** -US\$6.90bn (P: -US\$4.21bn)
- **Hungary September PPI:** 2.4% y/y (P: 2.3%)
- **Poland October CPI:** 2.8% y/y (MSe: 3.0%; C: 3.0%; P: 2.9%)
- **South Africa September Trade Balance:** ZAR21.8bn (C: ZAR6.9bn; P: ZAR4.0bn)
- **Brazil September National Unemployment Rate:** 5.6% (P: 5.6%; C: 5.5%). The flat print reflects labor market stabilization, consistent with muted consumer confidence and modest GDP growth.

### The Day Ahead

#### Developed Markets

- Sun 7:30pm ET - **Australia September Building Approvals** (MSe: 3.5% y/y, 5.0% m/m; C: 5.0% m/m; P: -6.0% m/m) & **Household Spending Indicator** (MSe: 5.4% y/y, 0.2% m/m; C: 5.5% y/y, 0.4% m/m; P: 5% y/y, 0.1% m/m)
- Mon 6:45am BST - **France Industrial Production** (MSe: 0.2% y/y; C: 0.1%; P: -0.7%)

- 8:50am BST - **ECB Governing Council member Simkus** will speak in Vilnius
- 9:30am GMT - **UK October Final Manufacturing PMI** (MSe: 49.6; C: 49.6; P: 49.6)
- 10:00am EST - **US October ISM Manufacturing** (MSe: 49.5 ; C: 49.4; P: 49.1). We forecast the ISM manufacturing composite at 49.5 for October, a half-point rise from September that leaves it barely above its averages of 48.6 in 3Q and 48.7 in 2Q. Actual manufacturing output swung from falling to rising gradually (~1%y/y) about a year ago, and there is no signal from the October factory surveys of any break in that uptrend.
- **US October Wards Total Vehicle Sales:** (MSe: 15.4m ; C: 15.5m; P: 16.4m). The electric vehicle tax credit ended on September 30. Demand pulled ahead of the expiration of the credit boosted sales in July-September, with higher EV penetration. We expect some reversal in October, and limited EV sales, bringing total sales to 15.4mn. This is down from the prior 3m average of 16.4mn.
- 12:00pm EST - **San Francisco Fed President Daly** will speak
- 2:00pm EST - **Fed Governor Cook** will speak about economic and monetary policy

#### Emerging Markets

- Sun - **Korea October Trade Balance** (MSe: \$4.07bn; C: \$3.0bn; P: \$9.5bn); **October Exports** (MSe: -4.1% y/y; C: -0.6%; P: 12.6%) and **Imports** (MSe: -6.0% y/y; C: -0.4%; P: 8.2%)
- Mon 8:30am HKT - **October S&P Global Manufacturing PMI: Indonesia** (P: 50.4); **Korea** (P: 50.7); **Malaysia** (P: 49.8); **Philippines** (P: 49.9); **Thailand** (P: 54.6); **Taiwan** (P: 46.8)
- 9:45am HKT - **China October RatingDog Manufacturing PMI** (C: 50.7; P: 51.2)
- 12:00pm HKT - **Indonesia October Headline CPI** (C: 2.6% y/y; P: 2.7%); **Core CPI** (C: 2.2% y/y; P: 2.2%); **September Trade Balance** (C: \$4.4bn; P: \$5.5bn); **September Exports** (C: 7.3% y/y; P: 5.8%) and **Imports** (C: 3.3% y/y; P: -6.6%)
- 1:00pm HKT - **India October Final HSBC Manufacturing PMI** (Preliminary: 58.4; P: 57.7)
- 9:00pm HKT - **Singapore October PMI** (P: 50.1)
- 7:00am BST - **Turkey October CPI** (MSe: 33.36% y/y; C: 33.20%; P: 33.29%)
- 7:00am BST - **Turkey October Manufacturing PMI** (P: 46.7)
- 8:00am BST - **Poland October Manufacturing PMI** (C: 48.6; P: 48.0)
- 8:00am BST - **Hungary October Manufacturing PMI** (C: 50.5; P: 51.5)
- 8:30am BST - **Czech October Manufacturing PMI** (C: 49.5; P: 49.2)

- 9:00am BST - **South Africa October Manufacturing PMI** (P: 52.2)
- 13:00pm BST - **Czech October Budget Balance** (P: -CZK153.9bn)
- 6:30am EST - **Chile September IMACEC** (P: 0.5% y/y; C: 3.2%).  
Consensus expects a sharp rebound in activity, driven by stronger mining and services output. The market is currently pricing in ~7bps of cuts for the remainder of 2025 and ~25bps of cuts in 2026, with a terminal rate near 4.43%.

For more details, please refer to our economists' weekly reports and data previews:

- [US](#)
- [Japan](#)
- [Europe](#)
- [Asia/Pacific](#)
- [LatAm](#)
- [CEEMEA](#)

Please [click here](#) if you would like to receive our weekly strategic advice on global rates, FX, and EM. See our latest [Global Macro Strategist: It Ain't What You Don't Know That Gets You Into Trouble...](#) and recent publications:

- US Rates Strategy: [ADP Is Dressing Up Like NFP For Halloween – Trick or Treat?](#)
- UK Rates Strategy: [November Budget Gilt Remit Projections](#)
- Euro Area Rates Strategy: [Add EU vs Germany to Overweight Periphery vs Core](#)

# G4 Smarter (Beta) Trading Partners

MORGAN STANLEY &amp; CO. LLC

**Matthew Hornbach, CMT**[Matthew.Hornbach@morganstanley.com](mailto:Matthew.Hornbach@morganstanley.com)

+1 212 761-1837

Since 2000, enhancements to a G4 10y government bond futures momentum strategy have produced higher Sharpe ratios and stronger returns relative to total return government bond indices for the G4: US, Germany, Japan, and the UK. See [A "Smarter" \(Beta\) Way to Trade G4 10y Futures Duration?](#) for more information on these strategies.

## Exhibit 5: Trading Signals for G4 Smarter (beta) Trading Strategy

| Current Risk, G4 10y Futures | G4 Strategy Weight | Trade Longs Portfolio | Fade Shorts Portfolio | Total Risk Trade Longs Only | Total Risk Trade Longs/Fade Shorts (max 200%) | Trade Longs Portfolio Entry Date | Trade Longs Portfolio Exit Date | Fade Shorts Portfolio Entry Date | Fade Shorts Portfolio Exit Date |
|------------------------------|--------------------|-----------------------|-----------------------|-----------------------------|-----------------------------------------------|----------------------------------|---------------------------------|----------------------------------|---------------------------------|
| J8 10y Future                | 32.50%             | 100%                  | 100%                  | 100%                        | 200%                                          | 10/15/2025                       | 11/20/2025                      | 9/19/2025                        | 10/28/2025                      |
| GE 10y Future                | 29.25%             | 100%                  | 0%                    | 100%                        | 100%                                          | 10/7/2025                        | 11/11/2025                      | -                                | -                               |
| US 10y Future                | 30.50%             | 100%                  | 100%                  | 100%                        | 200%                                          | 10/14/2025                       | 11/19/2025                      | 9/24/2025                        | 10/30/2025                      |
| UK 10y Future                | 7.75%              | 100%                  | 100%                  | 100%                        | 200%                                          | 10/15/2025                       | 11/19/2025                      | 9/25/2025                        | 10/30/2025                      |

Source: Morgan Stanley Research

### Trading Strategy 1 – "Trade Longs/Fade Shorts"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, buy the futures contract and hold for a 25-business-day period. In short, this strategy buys futures when the Simple Moving Average Crossover (SMAX) generates both a long and a short signal, given the historical outperformance of long signals traded long and the underperformance of short signals traded short. Given that the SMAX could generate both a long and a short signal within the predefined holding period, an investor may have a 200% long position since each of the two signals would be traded in separate portfolio sleeves.

### Trading Strategy 2 – Trade "Longs Only"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, do nothing. In short, an investor *only* trades long signals initiated by the SMAX given their historical precedent to outperform.



# US Holiday Trading Strategy

## Currently Active Strategies

None.

Combining the individual trading signals based on 10y Treasury future (TY) price patterns before and after US holidays has generated an annual strategy with no down years since 1987. See [Trading 10y Treasury Futures Around US Holidays](#) for more information.

Using historical data since 1983, we created a trading strategy using 10y Treasury futures (TY) that trades 11 times per year, either long or short, before or after the following US holidays: New Year's Day, Martin Luther King Day, Presidents' Day, Good Friday, Memorial Day, Independence Day, Labor Day, Columbus Day, Thanksgiving, and Christmas. [Exhibit 6](#) displays the trading rules:

**Exhibit 6:** Trading rules for US holiday trading model using US 10y futures contracts (TY)

| Holiday                     | Day of Week | Direction | Before/After Holiday | Trading Rule (business days, 3pm NY unless specified)                  | Trade Entry (business days) | Trade Exit (business days) |
|-----------------------------|-------------|-----------|----------------------|------------------------------------------------------------------------|-----------------------------|----------------------------|
| New Year's Day              | Rotates     | Long TY   | After                | Buy TY New Years +1, Sell TY New Years +4                              | New Year + 1                | New Year + 4               |
| Martin Luther King Day 1    | Monday      | Long TY   | Before               | Buy TY Tuesday before MLK, Sell Friday before MLK                      | MLK - 4                     | MLK - 1                    |
| Martin Luther King Day 2    | Monday      | Short TY  | After                | Sell TY Tuesday after MLK, Buy Friday after MLK                        | MLK + 1                     | MLK + 4                    |
| President's Day             | Monday      | Long TY   | After                | Buy TY Tuesday after Pres. Day, Sell TY Friday after Pres. Day         | Pres. Day + 1               | Pres. Day + 4              |
| Good Friday                 | Friday      | Long TY   | After                | Buy TY Monday after Good Friday, Sell TY Friday after Good Friday      | Good Friday + 1             | Good Friday + 5            |
| Memorial Day                | Monday      | Long TY   | After                | Buy TY Tuesday after Mem. Day, Sell TY Friday after Mem. Day           | Mem. Day + 1                | Mem. Day + 4               |
| Independence Day (July 4th) | Rotates     | Long TY   | After                | Buy TY Independence Day + 2, Sell TY Independence Day + 4              | Ind. Day + 2                | Ind. Day + 4               |
| Labor Day                   | Monday      | Long TY   | Before               | Buy TY Monday before Labor Day, Sell TY Thursday before Labor Day      | Labor Day - 4               | Labor Day - 2              |
| Columbus Day                | Monday      | Short TY  | Before               | Sell TY Monday before Col. Day, Buy TY Thursday before Col. Day        | Colum. Day - 5              | Colum. Day - 2             |
| Thanksgiving                | Thursday    | Long TY   | Before               | Buy TY Friday before Thanksgiving, Sell TY Tuesday before Thanksgiving | Thanksgiving - 4            | Thanksgiving - 2           |
| Christmas                   | Rotates     | Short TY  | Before               | Sell TY Christmas Day - 4, Buy TY Christmas Day - 1                    | Christmas - 4               | Christmas - 1              |

Source: Morgan Stanley Research. Note: Past performance is no guarantee of future results.

# Global Macro Strategy Team

|                                          |                                                                                                                            |                                                                     |                  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------|
| MORGAN STANLEY & CO. LLC                 | <b>Matthew Hornbach, CMT</b><br><a href="mailto:Matthew.Hornbach@morganstanley.com">Matthew.Hornbach@morganstanley.com</a> | Global Head of Macro Strategy                                       | +1 212 761-1837  |
|                                          | <b>Bradley Tian</b>                                                                                                        | Global Macro Commentary                                             | +1 212 761-0826  |
|                                          | <b>Martin Tobias, CFA, CMT</b>                                                                                             | US Rates Strategist                                                 | +1 212 761-6076  |
|                                          | <b>Shaun Zhou</b>                                                                                                          | US Rates Strategist                                                 | +1 212 761-3348  |
|                                          | <b>Aryaman Singh</b>                                                                                                       | US Rates Strategist                                                 | +1 212 761-1993  |
| MORGAN STANLEY & CO. LLC                 | <b>Eli Carter</b>                                                                                                          | US Rates Strategist                                                 | +1 212 761-4703  |
|                                          | <b>Andrew Watrous</b>                                                                                                      | G10 FX Strategist                                                   | +1 212 761-5287  |
|                                          | <b>Molly Nickolin</b>                                                                                                      | G10 FX Strategist                                                   | +1 212 761-3592  |
|                                          | <b>Simon Waever</b><br><a href="mailto:Simon.Waever@morganstanley.com">Simon.Waever@morganstanley.com</a>                  | Head of EM Sovereign Credit and Latin America Fixed Income Strategy | +1 212 296-8101  |
|                                          | <b>Ioana Zamfir</b>                                                                                                        | Latin America Macro Strategist                                      | +1 212 761-4012  |
| MORGAN STANLEY & CO. INTERNATIONAL PLC   | <b>Emma Cerda</b>                                                                                                          | Latin America Sovereign Credit                                      | +1 212 761-2344  |
|                                          | <b>James K. Lord</b><br><a href="mailto:James.Lord@morganstanley.com">James.Lord@morganstanley.com</a>                     | Global Head of FXEM Strategy                                        | +44 20 7677-3254 |
|                                          | <b>Gianluca Salford</b><br><a href="mailto:Luca.Salford@morganstanley.com">Luca.Salford@morganstanley.com</a>              | Head of European Rates Strategy                                     | +44 20 7677-1337 |
|                                          | <b>Lorenzo Testa</b>                                                                                                       | Euro Area Rates Strategist                                          | +44 20 7677-0337 |
|                                          | <b>Fabio Bassanin, CFA</b>                                                                                                 | UK Rates Strategist                                                 | +44 20 7425-1869 |
| MORGAN STANLEY ASIA LIMITED+             | <b>Maria Chiara Russo</b>                                                                                                  | Euro Area Rates Strategist                                          | +44 20 7677-3499 |
|                                          | <b>David S. Adams, CFA</b><br><a href="mailto:David.S.Adams@morganstanley.com">David.S.Adams@morganstanley.com</a>         | Head of G10 FX Strategy                                             | +44 20 7425-3518 |
|                                          | <b>Neville Mandimika</b>                                                                                                   | CEEMEA Sovereign Credit Strategist                                  | +44 20 7425-2509 |
|                                          | <b>Arnav Gupta</b>                                                                                                         | CEEMEA Rates Strategist                                             | +44 20 7677-0382 |
|                                          | <b>David Cueva Oemer</b>                                                                                                   | CEEMEA Rates Strategist                                             | +44 20 7425-2143 |
| MORGAN STANLEY MUFG SECURITIES CO., LTD. | <b>Gek Teng Khoo</b>                                                                                                       | AXJ Macro Strategist                                                | +852 3963-0303   |
| MORGAN STANLEY MUFG SECURITIES CO., LTD. | <b>Koichi Sugisaki</b><br><a href="mailto:Koichi.Sugisaki@morganstanleymufg.com">Koichi.Sugisaki@morganstanleymufg.com</a> | Head of Japan Macro Strategy                                        | +81 3 6836-8428  |
|                                          | <b>Hiromu Uezato</b>                                                                                                       | Japan Macro Strategist                                              | +81 3 6836-8431  |

## Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at [www.morganstanley.com/researchdisclosures](http://www.morganstanley.com/researchdisclosures), or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

### Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David S. Adams, CFA; Fabio Bassanin, CFA; Eli P. Carter; David Cueva; Arnab Gupta; Matthew Hornbach; Gek Teng Khoo; James K. Lord; Molly Nickolin; Sofia Palacios; Nimish M. Prabhune; Maria Chiara Russo; Luca Salford; Aryaman Singh; Koichi Sugisaki; Lorenzo Testa; Bradley Tian; Martin W. Tobias, CFA; Hiromu Uezato; Andrew M. Watrous; Ioana Zamfir; Shaun Zhou.

### Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at [www.morganstanley.com/institutional/research/conflict/policies](http://www.morganstanley.com/institutional/research/conflict/policies). A Portuguese version of the policy can be found at [www.morganstanley.com.br](http://www.morganstanley.com.br)

### Important Regulatory Disclosures on Subject Companies

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Germany.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Germany.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Germany.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Germany.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

### STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

### Global Stock Ratings Distribution

(as of October 31, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1501              | 41%        | 384                              | 46%            | 26%                  | 703                                               | 41%                   |
| Equal-weight/Hold     | 1609              | 44%        | 370                              | 44%            | 23%                  | 788                                               | 46%                   |
| Not-Rated/Hold        | 4                 | 0%         | 1                                | 0%             | 25%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 568               | 15%        | 80                               | 10%            | 14%                  | 223                                               | 13%                   |
| Total                 | 3,682             |            | 835                              |                |                      | 1715                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

## Analyst Stock Ratings

**Overweight (O or Over)** - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

**Equal-weight (E or Equal)** - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

**Not-Rated (NR)** - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

**Underweight (U or Under)** - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

## Analyst Industry Views

**Attractive (A):** The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

**In-Line (I):** The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

**Cautious (C):** The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

## Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at [www.morganstanley.com/online/researchdisclosures](http://www.morganstanley.com/online/researchdisclosures). For Morgan Stanley specific disclosures, you may refer to [www.morganstanley.com/researchdisclosures](http://www.morganstanley.com/researchdisclosures).

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

## Other Important Disclosures

Morgan Stanley & Co. International PLC and its affiliates have a significant financial interest in the debt securities of Germany, Japan, United Kingdom.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy ([http://www.morganstanley.com/privacy\\_pledge.html](http://www.morganstanley.com/privacy_pledge.html)), including for the purposes of

setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy ([http://www.morganstanley.com/privacy\\_pledge.html](http://www.morganstanley.com/privacy_pledge.html)). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited ([https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms\\_and\\_conditions.pdf](https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf)) and the following link to review the audit report ([https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Research\\_Audit\\_Report.pdf](https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Research_Audit_Report.pdf)).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The 'Important Regulatory Disclosures on Subject Companies' section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: [tejarshi.hardas@morganstanley.com](mailto:tejarshi.hardas@morganstanley.com); Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: [msic-compliance@morganstanley.com](mailto:msic-compliance@morganstanley.com). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct

Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products. Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Shaun Zhou; Molly Nickolin; David S. Adams, CFA; Eli P Carter; Koichi Sugisaki; James K Lord; Aryaman Singh; Bradley Tian; Matthew Hornbach; Nimish M Prabhune; Ioana Zamfir; Gek Teng Khoo; Martin W Tobias, CFA; Andrew M Watrous; Lorenzo Testa; Luca Salford; Fabio Bassanin, CFA; Maria Chiara Russo; Hiromu Uezato; David Cueva.

The following authors are neither Equity Research Analysts/Strategists nor Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity or fixed income securities: Lingdi Xu.